SITUATION AWARENESS

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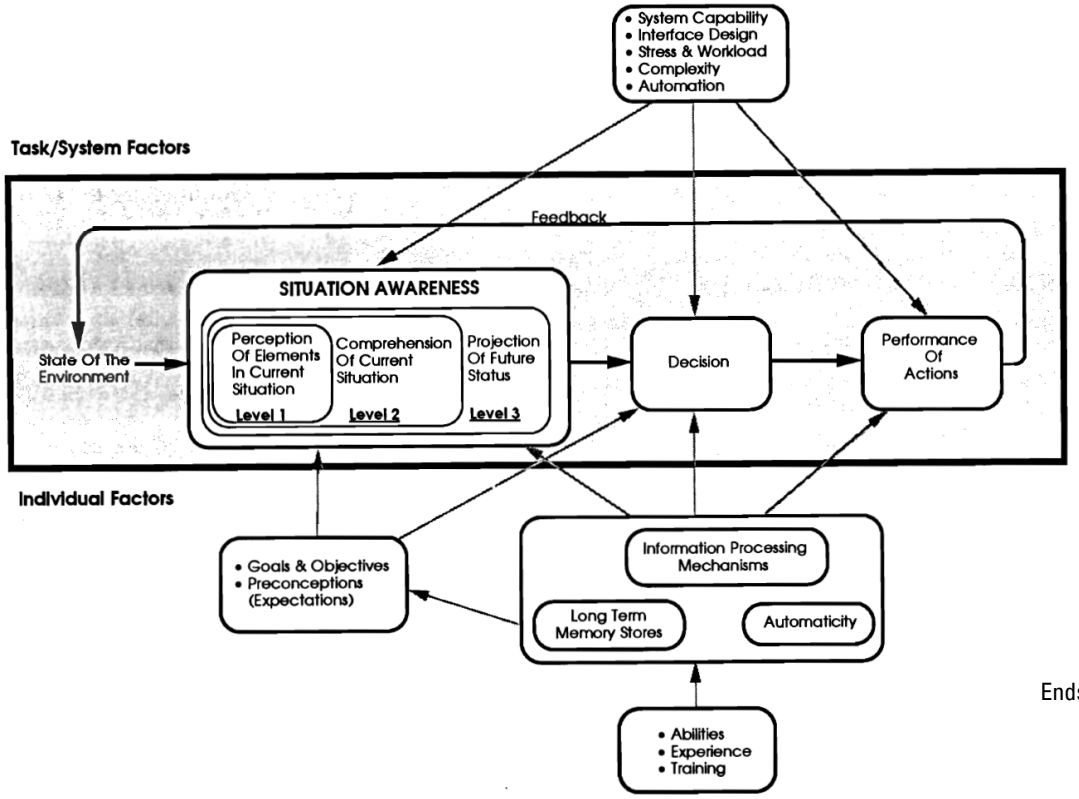
Credit: Gary Larson, The Far Side

SITUATION AWARENESS

"Situation awareness is the ***perception*** of the elements in the environment within a volume of time and space, the ***comprehension*** of their meaning, and the ***projection*** of their status in the near future." -- Endsley (1995)

- Perception (Level 1): Recognizing the status, attributes and dynamics of relevant elements in the environment.
- Comprehension (Level 2): Understanding the significance of the relevant elements in the context of operator goals.
- Projection (Level 3): Projecting the future actions of the elements in the environment while accounting for their status and dynamics.

SITUATION AWARENESS IN DECISION MAKING



Endsley, 1995

EXAMPLE OF SA ELEMENTS: DRIVING A CAR

Perception (Level 1) **Comprehension (Level 2)** **Projection (Level 3)**

BREAKDOWNS IN SA

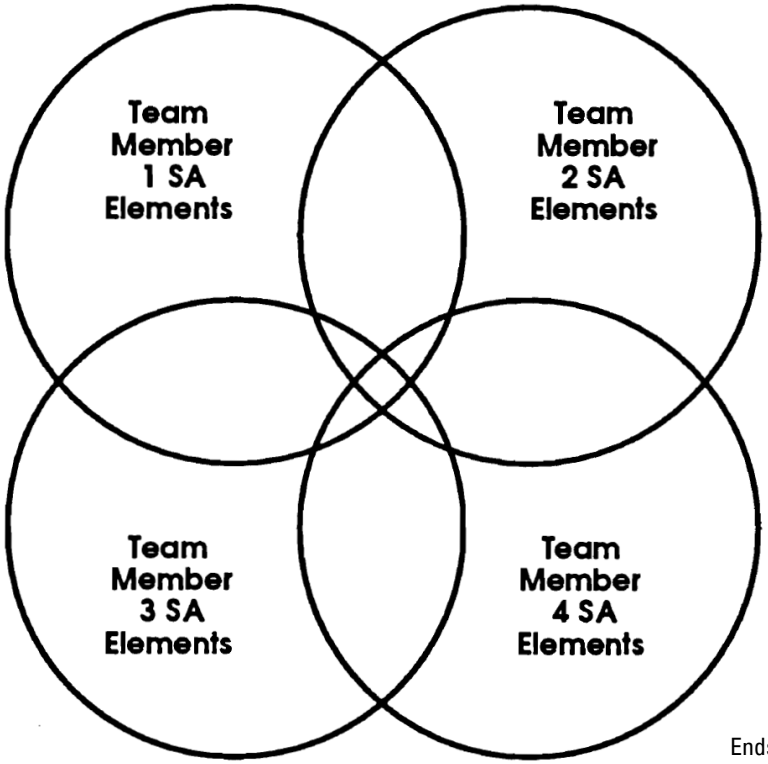
- Perception (Level 1):
  - Lack of detectability or distinguishability
  - Lack of attention
  - Visual obstructions, auditory masking
  - …
- Comprehension (Level 2):
  - No mental model for environment/observations
  - Incorrect mental model used to interpret perceived data
  - Data inappropriately used (e.g., auditory alarms that can have multiple meanings)
  - …
- Projection (Level 3):
  - Lack of a good enough mental model
  - Limited attentional or memory resources to perform projection
  - …

A breakdown in a level is when you do not have the relevant output from the SA level

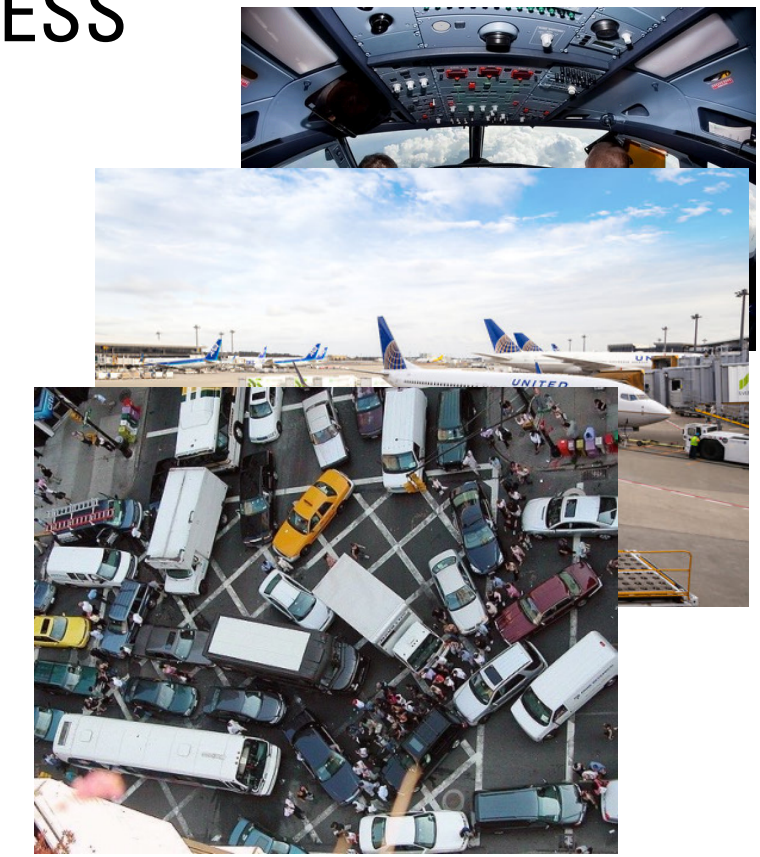
AMTRAK ACCIDENT REVISITED

- Where did the breakdown in situation awareness occur?
  - Level 1 – Perception
  - Level 2 – Comprehension
  - Level 3 - Projection

TEAM SITUATION AWARENESS



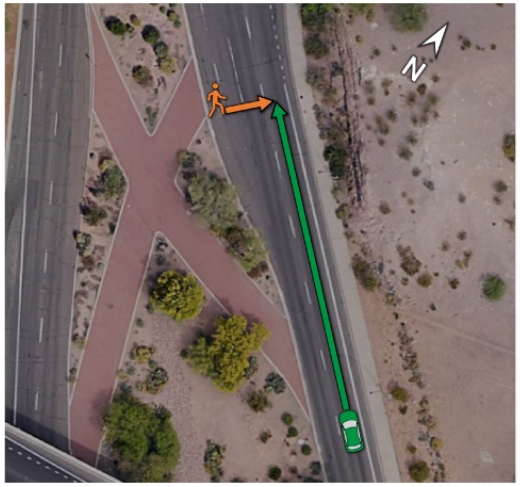
Endsley, 1995

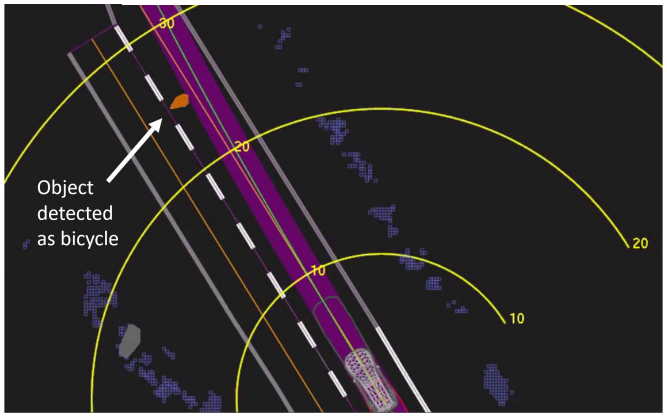


UBER ACCIDENT, MARCH 2018

NTSB/HAR-19/03

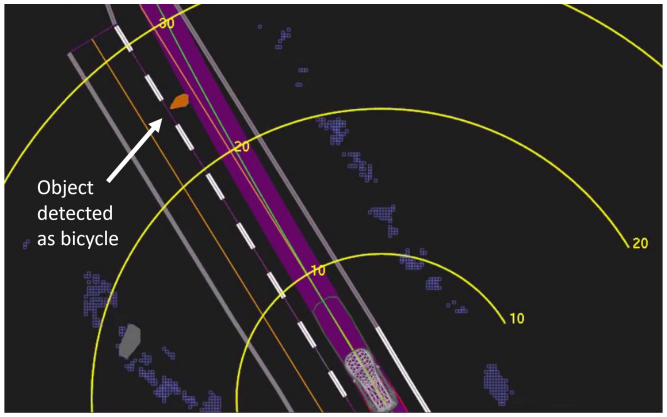
- Operator took self-driving car out of garage to begin evaluations at 9:14 pm
- Self-driving system in computer control mode struck a pedestrian at 9:58 pm
- Human operator intervened <1 sec before impact (engaged the steering wheel)
- The vehicle speed at impact was 39 mph.
- Operator began braking less than a second <u>after</u> the impact.
- The data showed that all aspects of the self-driving system were operating normally at the time of the crash, and that there were no faults or diagnostic messages.





UBER ACCIDENT, MARCH 2018

- Software classified the pedestrian as an unknown object, vehicle, and bicycle with varying probabilities and expectations of future travel path (first registered 6 seconds prior to impact).
- 1.3 seconds before impact, the software determined that an emergency braking maneuver was needed to mitigate a collision
- Emergency braking maneuvers are not enabled while the vehicle is under computer control, to reduce the potential for erratic vehicle behavior.
- The vehicle relies on the operator to intervene and take action.



Video released by Uber

DESIGN AN INTERFACE FOR TRANSITIONS

SUV interior

- Locations that could mount a cell phone (yellow region in center console)
- ADS engagement/disengagement knob (red)
- ADS engagement button (blue)
- HMI (green)
  - inset illustrating image on tablet.

EXPERIMENTAL MEASURES OF SA

- **Explicit:** Questions are defined that specifically ask operators about their SA
- **Implicit:** Responses are used to infer an operator's SA
- **Subjective:** Measures that are provided by the operator self-assessments or an observer
- **Objective?** What objective measures could we use? What are the pitfalls?

EXPLICIT MEASURES OF SA

Asks operators questions about their perception of the state of the system/environment

- Retrospective
  - Time dependent
  - Are you testing memory or SA?
- Concurrent (Situation Presence Assessment Method, SPAM) (Durso et al., 1995)
  - Workload may be high
  - Adding another task
- Freeze (SA Global Assessment Technology, SAGAT) (Endsley et al., 1988)
  - Intrusive

IMPLICIT MEASURES OF SA

Experimenter <u>infers</u> SA from performance-based measures

- Advantages
  - Objective and unobtrusive
  - Can be through embedded task measures
- Disadvantages
  - Poor performance isn't necessarily a function of low SA

SUBJECTIVE MEASURES OF SA

- Self-assessment or observer assessment
  - Situation Awareness Rating Technique (SART)
    - Instability - Likelihood of situation changing suddenly
  - Observer with system knowledge rates user's performance

- Advantages
  - Doesn't rely on memory
  - Non-intrusive
- Disadvantages
  - Are you testing confidence or SA?
  - Rating workload or SA?
  - Observer only has limited idea of the subject's internal knowledge states

		LOW						HIGH
		1	2	3	4	5	6	7
Demand	Instability of Situation							
	Variability of Situation							
	Complexity of Situation							
Supply	Arousal							
	Spare Mental Capacity							
	Concentration							
	Division of Attention							
Under	Information Quantity							
	Information Quality							
	Familiarity							

FIGURE 3.4
SART scale.

SA = Understanding – (Demand – Supply)

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